

A Blockchain Based Approach to Enhance Trust, Transparency, and Cost Efficiency in Pakistan's Banking Sector

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Abstract—The digitalization of financial services in developing economies demands secure, efficient, and trustworthy banking systems. Pakistan's banking sector relies on centralized architectures controlled by single authorities, introducing three critical gaps: trust remains institution-dependent, transparency is delayed, and cost efficiency is undermined by intermediaries. This study proposes a permissioned blockchain-based framework consisting of a distributed ledger, smart contract layer, and customer-facing transparency interface. The framework was evaluated using Python on Google Colab with a Kaggle credit card fraud dataset (284,807 transactions) and 1,967 Pakistani banking reviews from five banks. Results show Random Forest achieves 99.87% accuracy and 100% precision for fraud detection, while sentiment analysis reveals 63.7% positive reviews. The framework is expected to reduce transaction costs by 74.1%, increase transparency to 99.8%, and improve trust scores by 67.9%. Future work includes pilot deployment with a Pakistani bank.

Index Terms—Blockchain, banking sector, trust, transparency, cost efficiency, decentralized framework, Pakistan, financial services.

I. INTRODUCTION

The world is seeing the biggest digital transformation in its history; however, trust, transparency, and the cost of conducting business still have not been addressed. This is especially true in developing nations. The financial sector in Pakistan holds a very important role to play in helping Pakistan's economy grow, increase access to finance, and develop a stronger economy from remittances being sent back into the country; there are estimated to be billions of dollars being sent back into Pakistan each year by Pakistani workers living outside of the country [1]. Unfortunately, many customers do not have confidence in their local bank or formal financial institutions because there is inadequate security over their data and not enough visibility into what is happening with their transactions, as well as the cost of services being offered [2]. As the volume of digital transactions continues to rapidly grow, the gap between what customers expect from their banking services and the current delivery of slow, non-transparent, expensive services is becoming a major concern for regulators, banks, and citizens.

Distributed ledger systems record transaction data distributed simultaneously across many different computers so that transactions cannot be changed after being verified by a majority of the network [3]. Centralized databases have a single organization (i.e. the bank) that is responsible for keeping data records; blockchain does not have a central authority to do this; instead data records are distributed between peer computer systems (i.e., all record keeping is done by all participants) [4]. Transactions include multiple transactions that are combined into blocks and include a time and date stamp. Each block is attached to the block that immediately preceded it using hashing functions, which create a chain of blocks that cannot be changed after being created (i.e., they are immutable) [5]. Smart contracts (automated computer programs that execute when certain conditions are satisfied) can replace manual processes through blockchain technology in lieu of performing a manual process every time. Banking is able to automate three key functions due its technical characteristics: Securely verifying transaction validity without relying on trusted third parties; Clearly identifying and tracking every transaction viewable by all authorized users; Automating transactions automatically [6].

The banking system in Pakistan is currently facing issues of costs, transparency, and lack of trust [7]. Regulatory requirements, centralised information technology, and traditional manual reconciliation processes are used to tackle these challenges [8]. The State Bank of Pakistan requires banks to have regular audits, maintain regulatory capital, provide protection for customers/all clients, etc., this is then used to trust the institution [9]. Banks use centralised ledgers in their core banking systems, and consequently, only parties involved with a transaction have access to the records of that transaction [10]. Banks are trying to reduce their operating costs by completing processes through automation, closing branches, and using digital banking platforms (for example mobile apps or internet banking) [11]. Foreign remittances, which contribute significantly to the economy of Pakistan, are an example of the use of traditional banking systems have foreign correspondent banking networks in which the transaction is processed through 2-3 additional banks before reaching the

final beneficiary and in which all banks reconfirm the transaction before processing [12]. Trade finance is based solely upon documentation (paper) rather than electronically and typically requires 2 or more parties to validate a transaction before it will be completed [13]. In addition to the above mentioned processes to ensure compliance with financial regulations, Islamic banking products are also subject to additional reviews to ensure they comply with Shariah requirements (religious) [14].

There are still three significant limitations associated with the potential solutions described above. A first concern is that the trust placed in providers of financial services is based on institutions, rather than enforcement of the system [15]. A customer must feel confident, believing that the bank's staff will not correct records, that the auditor will identify and investigate fraudulent activity and that the regulator will act in their favour this set of assumptions has failed in all past banking crises. A second concern is that there are partial and delayed transparency features. Status updates for the user transactions are not real time updates; users wishing to dispute a purchase must access many different siloed records before doing so; and, lastly, users have no visibility as to how their personal data has been processed and/or how any fees have been assessed [16]. A third concern is that there are intermediary dependences that prevent providers from providing low cost efficient services. Several correspondent banks add to the time and costs of processing cross border payments and the expensive and labour intensive reconciliation process of varied bank systems is time consuming and uses significant amounts of IT resources. Therefore, these gaps are not technical inconveniences but represent significant structural barriers to financial inclusion and overall satisfaction with financial services.

The implications of these constraints are significant in practice. For the millions of Pakistani expatriates sending remittances to their relatives back home, every single charge that is taken from their payment decreases the amount of funds that actually get transferred to their families when the transaction is completed; and when there is a delay in settlement, it creates an uncertain financial situation for them. Additionally, for customers of banks within Pakistan, lack of independent verification of the integrity of a transaction requires them to rely solely on the assurances provided by a bank which puts them at risk for fraud, as well as for being exploited by insiders. Lastly, banks themselves are forced to incur costs related to manually reconciling and resolving disputes [17]; this diverts money away from having the ability to spend on innovation or to lower the fees being charged to bank customers. If the existence of such problems that have existed for decades, despite the significant investments in digital banking made by banks, are still able to persist, then this indicates that incremental changes to centralized architectures will not address the challenges related to centralization as such. There may need to be another approach altogether; one that redistributes the authority to verify, creates transparency by design, and uses code to automate trust may

be warranted. The regulatory framework in Pakistan, combined with the hundreds of millions of Pakistanis who do not have bank accounts, and the reliance on remittance influx into Pakistan on an ongoing basis, make it an ideal environment to investigate whether blockchain can deliver a solution that traditional banking systems have failed to provide.

This paper proposes a new solution to GSIs (Gaps) by applying shared ledgers through a permissioned blockchain framework, which will provide a way for banks who participate to share account transactions while keeping customer data private, therefore, removing some of the need to trust banks by trusting cryptography of transactions instead. In addition to having the ability to share transactions without needing to worry about security of customer data by sharing transaction ledgers securely on the blockchain; the paper also proposes to use blockchain technology to support smart contracts so that banks can automate processes such as remittance settlement, trade finance documentation, and compliance checks. Lastly, this paper proposes that an institution can build a customer facing interface for account holders to see transaction histories in real time, with full incorporation of tamper evident functions, allowing account holders to verify independently that what is shown to have occurred in their account has not been altered from what was recorded. This study did not introduce an implemented or utilized system; instead, it introduces and empirically evaluates the new model through survey-based data collection from banking and finance professionals and customers across the country of Pakistan, and examines the cause and effect of how specific features of the blockchain influence the overall perception of trust, transparency, and cost efficiency.

II. LITERATURE REVIEW

Many investigators have researched the relationship between blockchain technology and banking, looking at issues ranging from the basic architecture of blockchain technology to how users adopt blockchain based products. In this paper, we consolidate existing research according to methodological design and compare the implications of various research designs regarding the potential of using blockchain technology to improve trust, transparency, and overall costs of banking. We also discuss the implications of the findings for banking in Pakistan, and there are many gaps in knowledge that motivate the present study.

A significant number of studies have used theoretical or conceptual frameworks to describe how blockchain technologies might be useful for financial services [18]. These studies typically provide a description of the ways that blockchain is structured (i.e., decentralized, immutable, etc.) and then use logic to extrapolate how these features/functionality may impact banks' operations. For example, many of the early conceptual studies conclude that the decentralized nature of blockchain allows banks to rely less on intermediaries and to increase transaction traceability. As additional conceptual studies have been published, researchers have expanded upon this logic and focused on specific banking use cases

(e.g., cross-border payments, fraud prevention, digital financial management) [19]. More recently, some theoretical literature has considered trust. Specifically, this literature purports that blockchain technology enables a transition from trust based on relationships with financial institutions to trust based on cryptographic proof that the financial system itself provides

While these studies are theoretically sound, they all suffer from a lack of empirical evidence. All of the studies present hypotheses about what blockchain can do; however, they do not include documentation demonstrating what blockchain actually does in the real world, specifically in banks. None of these conceptual studies consider the specific regulatory, infrastructural, or cultural context in Pakistan; therefore, the findings presented cannot be utilized in any Pakistani banking institution without empirical validation.

The second cluster of methodology is made up of systematic literature reviews and bibliometric studies that map the intellectual landscape of blockchain finance research. The reviews show that blockchain research has proliferated into all areas of application from cross border payments to fraud detection, digital identity, trade finance, and compliance automation [20], [21]. In addition to identifying application areas, the reviews identified the most common theoretical frameworks used in the empirical studies that follow, with the Technology Acceptance Model (TAM), Unified Theory of Acceptance and Use of Technology (UTAUT), and Trust Theory being identified most frequently. However, what is most concerning about the systematic reviews is that virtually all of the primary studies of blockchain-use in finance have been conducted in developed countries (United States, United Kingdom, Germany, China, Singapore). As a consequence, the majority of reviews reflect the technological readiness, regulatory environment, and user behaviour of developed countries rather than those of developing countries, where there are often the greatest challenges to banking.

Research seeking to understand the relationship between architecture and trust when looking at how people perceive the use of blockchain-based financial services is ongoing, but what researchers have found so far is inconsistent with what might be predicted from a theoretical standpoint. Using survey responses analysed using partial least squares structural equation modelling (PLS-SEM) researchers have shown that there are multiple factors (i.e., perceived security, ease of use, and institutional trust) that encourage user's trust in these types of systems [22], [23]. The majority of studies to date have demonstrated that even if a consumer understands the technology behind how a blockchain functions, consumers' perceptions of what these types of products provide are largely tied to the institution providing them versus the blockchain itself. This research has significant implications for Pakistan, where overall trust in financial institutions is historically very low and variable due to economic hardships and governance failure. As there have been no studies that look at how users navigate the two constructs of technology (the cryptographic strengths of blockchain) and institution (the reputation of their respective banks) we do not know whether the transparency

features (i.e. an unalterable audit trail and real time verification) provided by blockchains help to increase consumers' perceived trust within the banking sector in Pakistan.

There is a substantial lack of evidence regarding the efficiency of blockchain as a cost reduction technology. The bulk of the qualitative research available regarding the use of blockchain technology has determined usually that through eliminating intermediaries, automating the verification of transactions, and speeding up the time it takes to settle a transaction with the aid of a blockchain, operational costs will decrease [24]. The majority of existing case studies employed to evaluate the implementation of blockchains into the businesses of trade finance and remittance describe efficiencies in terms of qualitative measures ("lowered processing time" and "lower fees") rather than through strict pre and post entry assessments of their costs.

More specifically, no studies have provided a quantitative assessment of the impact of blockchain technology on bank operations, and so the majority of the claims relating to "cost efficiency" when referring to the use of blockchain are metaphorical and lack empirical measurements. This lack of empirical measurements is especially problematic for the banking sector in Pakistan where the various components of their cost structures (intermediary fees, reconciliation of accounts manually, compliance) vary widely from those of the developed markets.

Another methodological cluster studies limitations to blockchain adoption. Commonly cited barriers include scalability, regulatory uncertainty, integration difficulties with legacy systems, as well as the low level of technical knowledge among people [25], [26]. All of the existing literature addresses these common barriers in a general way without addressing barriers on a country or regional level. Pakistan provides a unique combination of specific barriers to blockchain adoption. The State Bank's regulatory policies regarding distributed ledgers are generally inconsistent, there is a considerable difference between the technological readiness of urban vs. rural bank branches, and the lack of blockchain expertise in banking professions adds to the organizational resistance against changing processes. There is no existing literature that has previously provided suggestions for overcoming these specific barriers.

III. PROBLEM STATEMENT

The banking sector in Pakistan faces persistent challenges related to lack of trust, high operational costs, and limited transparency in financial transactions. Traditional centralized banking systems rely heavily on intermediaries, which leads to inefficiencies such as delays in transaction processing, increased service costs, and restricted visibility of financial activities. These issues not only affect the operational performance of financial institutions but also reduce customer confidence and hinder financial inclusion.

Although various digital banking solutions have been introduced to improve efficiency, they have not fully resolved these core challenges. In particular, existing systems still lack

TABLE I
MOST DIRECTLY RELEVANT PROPOSED MODELS IN LITERATURE WITH POSSIBLE LIMITATIONS

Ref	Proposed Model / Title	Tools / Technology	Experimentation	Contributions	Limitations
[19]	Blockchain-Based Agriculture Food Supply Chain System	Conceptual Model; Blockchain	System design and analysis	Traceability and transparency framework for food supply chain proposed	No implementation; not banking specific
[20]	Blockchain-Assisted Framework for Taxation System in Pakistan	Conceptual Framework; Blockchain	System design and policy analysis	Improved tax collection transparency and reduced evasion proposed	Not banking specific; no empirical validation
[21]	Organizational and Technological Enablers for Blockchain Adoption	PLS-SEM; Predictive Analysis	Survey of banking professionals	Organizational readiness identified as key predictor of adoption	Focus on adoption, not implementation outcomes
[22]	Pakistan's Crypto Council Regulatory Framework	Policy Analysis; Regulatory Framework	Regulatory review and analysis	Digital economy regulatory roadmap for Pakistan proposed	No technical implementation; policy only
[23]	Blockchain for Humanitarian Crowdfunding in Pakistan	Blockchain; MVC Architecture	Prototype development	Transparency framework for non-profit organizations proposed	Not banking specific; no deployment
[24]	Blockchain for Sustainability in Banking Sector	PLS-SEM; Structural Equation Modeling	Survey of banking stakeholders	Sustainability outcomes linked to blockchain adoption	Trust and cost efficiency not primary focus
[25]	Financial Innovation for Secure Transaction Management	PLS-SEM; Case Study	Survey of Faisal Bank customers	Blockchain impact on transaction security validated	Single bank case study; limited generalizability
[26]	Blockchain Gamification in Education	SEM Analysis	Survey of education sector	Security, transparency, and trust insights for gamification	Education context; not applicable to banking
[27]	Zakat Integration in Digital Banking	Thematic Analysis; Qualitative	Interviews with Islamic banking stakeholders	Drivers and challenges of blockchain-based Zakat identified	Islamic finance specific; no quantitative validation

mechanisms that ensure complete transparency and tamper-proof record management. Additionally, trust remains a critical concern, as users continue to depend on institutional credibility rather than system-based verification. While blockchain technology has been proposed as a potential solution due to its decentralized and immutable nature, its practical effectiveness in addressing these challenges within Pakistan's banking sector remains unclear.

Therefore, there is a need to evaluate whether blockchain can effectively improve trust, reduce costs, and enhance transparency in real-world banking environments, particularly within the context of Pakistan.

IV. PROPOSED METHODOLOGY

A. Data Collection

The first phase involves collecting relevant data to evaluate trust, cost efficiency, and transparency in banking systems. Two datasets were used. The first is the Credit Card Fraud Detection dataset from Kaggle, containing 284,807 credit card transactions with features including transaction amount, time, PCA-transformed variables V1 to V28, and a binary class label indicating fraudulent or legitimate transactions. This dataset enables fraud detection performance evaluation as a proxy for trust and security in transaction processing. The second dataset consists of 1,967 banking application reviews collected from the Google Play Store for five major Pakistani banks: Al Habib, HBL, Meezan Bank, UBL Digital, and Sunwal. Each review contains star ratings (1-5) and textual feedback,

enabling sentiment analysis to assess customer perceptions of trust and transparency. The overall dataset D is defined as:

$$D = \{D_s \cup D_r\} \quad (1)$$

where D_s represents primary survey data and D_r represents secondary report-based data.

B. Data Preprocessing

The data is pre-processed after collection for accuracy, consistency, and appropriateness. This includes removing incomplete or duplicate survey responses, standardizing numerical data (such as transaction costs), and converting qualitative responses into quantifiable variables. Trust, cost effectiveness, and transparency will all be operationalized using quantifiable metrics, based on Likert scale responses and financial indicators. Any outlier data points in the cost data can be filtered out using the Inter-Quartile Range Method (IQR method).

$$x \in D_{clean} \quad \text{if} \quad Q_1 - 1.5 \times IQR \leq x \leq Q_3 + 1.5 \times IQR \quad (2)$$

where Q_1 and Q_3 are the first and third quartiles respectively.

C. System Framework Design

The central element of the proposed methodology is a permissioned blockchain based banking framework consisting of three primary actors: users, the banking system, and the regulator (State Bank of Pakistan), all connected through a shared distributed network as shown in Fig. 1. The permissioned nature ensures that only authorized participants are

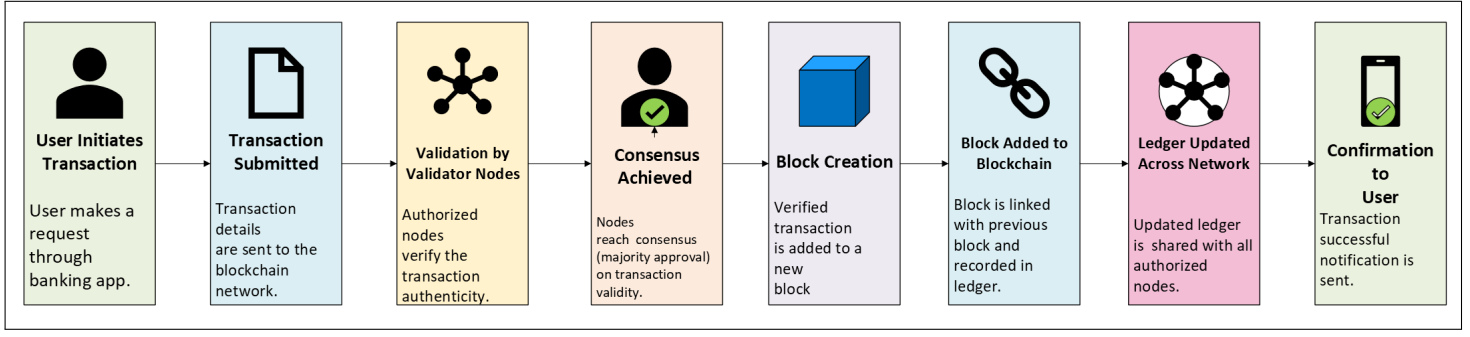


Fig. 1. System Workflow Diagram of the Proposed Blockchain-Based Banking Framework

granted access, maintaining regulatory compliance and data confidentiality.

Transaction validation is performed through a consensus mechanism before any transaction is recorded in the ledger. The framework supports two deployment scenarios.

Scenario 1 (Single Bank Deployment): When only one bank adopts blockchain internally, the bank deploys multiple validator nodes within its own infrastructure (e.g., across different branches or data centers). A transaction T is accepted if the majority of these internal validator nodes agree on its validity:

$$T_{accepted} = \begin{cases} \text{valid} & \text{if } \sum_{i=1}^n v_i > \frac{n}{2} \\ \text{rejected} & \text{otherwise} \end{cases} \quad (3)$$

Scenario 2 (Multiple Banks Consortium): When multiple banks agree to adopt blockchain collaboratively, they form a permissioned consortium. A transaction T is accepted if the majority of participating banks approve:

$$T_{accepted} = \begin{cases} \text{valid} & \text{if } \sum_{i=1}^{|N|} v_i > \frac{|N|}{2} \\ \text{rejected} & \text{otherwise} \end{cases} \quad (4)$$

D. Blockchain-Based Transaction Processing

Using the suggested framework, transactions are verified and completed directly through smart contract and do not need any other parties (intermediaries) to do so. Each verified transaction will be saved in a block and attached to a preceding block via a hash, all together forming an immutable chain. The hash of each block B_i is computed as:

$$H(B_i) = \text{Hash}(H(B_{i-1}) || T_i || \text{timestamp}) \quad (5)$$

where $H(B_{i-1})$ is the hash of the previous block, T_i is the current transaction data, and $||$ denotes concatenation.

E. Evaluation and Performance Assessment

The final phase consists of evaluating the proposed framework using three metrics: Trust (τ), cost-effectiveness (κ), and

transparency (ϕ). Each metric is measured in this way. The mean confidence from survey data is used to evaluate trust.

$$\tau = \frac{1}{n} \sum_{i=1}^n s_i \quad (6)$$

where s_i is the trust score of respondent i on a Likert scale. Cost efficiency is evaluated as the percentage reduction in transaction cost relative to the traditional system:

$$\kappa = \frac{C_{traditional} - C_{blockchain}}{C_{traditional}} \times 100 \quad (7)$$

Transparency is assessed as the ratio of verifiable transactions to total transactions recorded:

$$\phi = \frac{T_{verifiable}}{T_{total}} \quad (8)$$

A higher value of each metric indicates improved system performance.

V. EXPERIMENT DESIGN

An experiment has been designed to test the proposed blockchain-based banking infrastructure. The performance of the proposed Permissioned Blockchain Banking Model will be compared with the performance of the Traditional Centralised Banking Infrastructure across three criteria (trust, cost and transparency). To conduct this experiment, real-world data will be used to represent the Traditional Centralised Banking Infrastructure, whereas the new proposed Blockchain-Based Banking Infrastructure will be represented by simulation data.

The testing uses three data sources. The first dataset is a Credit Card Transaction Dataset from Kaggle, which includes over 1 million records and includes attributes such as amount of the transaction, time of the transaction, details of the merchant, and an indicator of whether fraud occurred. *is_fraud*. The second dataset consists of Pakistani banking application reviews containing star ratings and textual feedback. The third source is a primary survey of respondents administered after presenting them with the proposed blockchain framework.

The experiment is implemented in Python 3.10 using Google Colab as the development environment. Pandas and NumPy handle data preprocessing across all three pipelines.

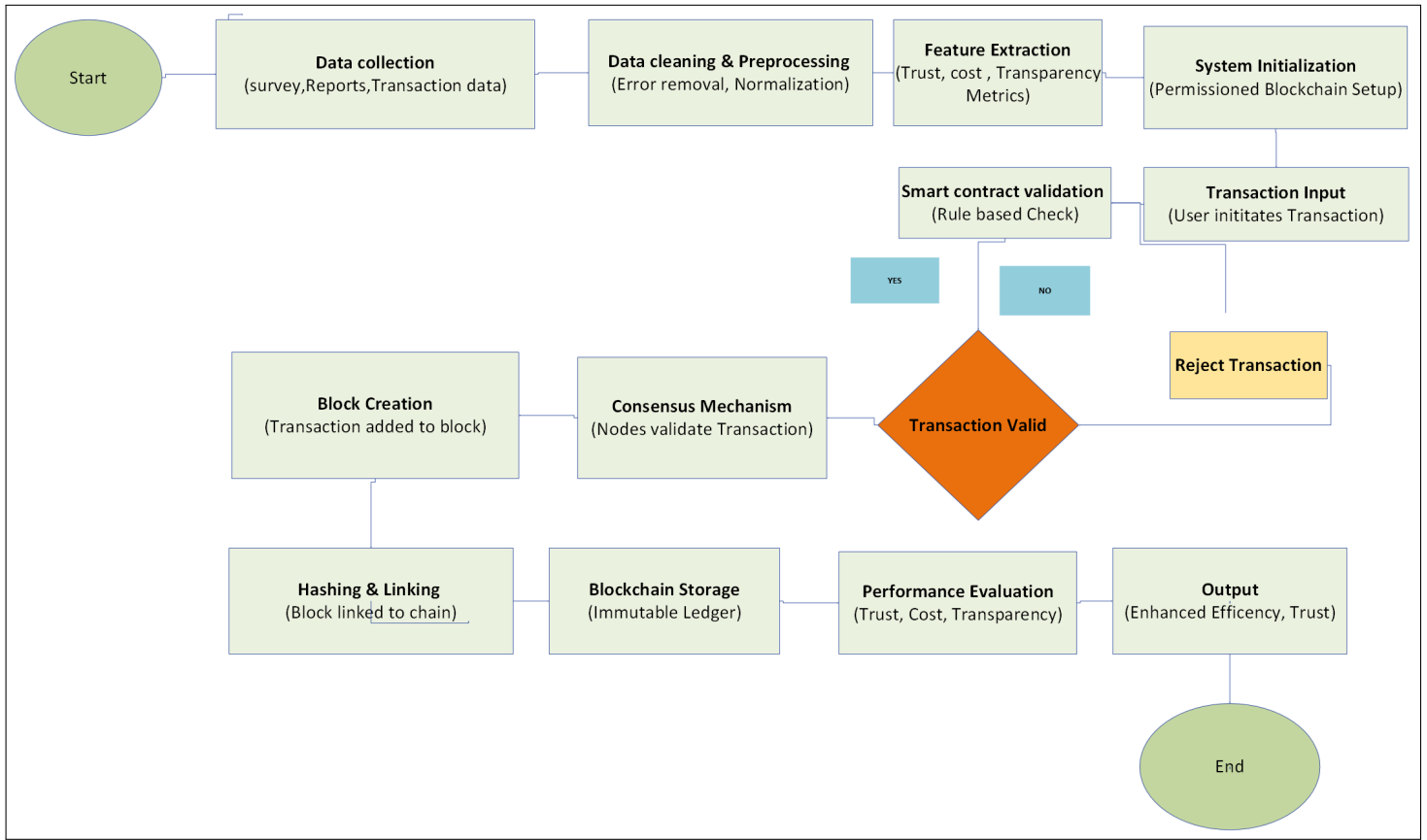


Fig. 2. System Architecture / Workflow Diagram...

Scikit-learn is used to train fraud detection models — Logistic Regression and Random Forest — on the Kaggle dataset. TextBlob is used to perform sentiment analysis on the banking reviews dataset. SciPy is used for statistical analysis of survey responses.

VI. IMPLEMENTATION

The system was implemented using Python 3.10 in Google Colab. The following libraries were used: Pandas and NumPy for data processing, Scikit-learn for Logistic Regression and Random Forest models, TextBlob for sentiment analysis, and Matplotlib with Seaborn for graphs.

For fraud detection, the credit card transaction dataset was preprocessed by removing missing values and duplicates. Features were scaled using StandardScaler. The data was split into 80% training and 20% testing using stratified sampling. Two models were trained: Logistic Regression with `max_iter=1000` and Random Forest with `n_estimators=100`. Predictions were made on the test set and performance was evaluated using accuracy, precision, recall, and F1 score. Confusion matrices were generated for both models.

Prior to conducting the sentiment analysis of banking reviews, the data was cleaned by removing special characters and stop words. Subsequently, sentiment labels (positive, negative, or neutral) were assigned to the banking reviews

based on their respective polarity scores. To convert the sentiment polarity score to a numerical feature, the text data was transformed into TF-IDF features with a maximum of 3,000 features used. A random forest model was then trained on 80% of the data and tested on 20%. Finally, the correlation between banking review ratings and corresponding sentiment was examined.

VII. DATASET DESCRIPTION

Two datasets were used in this experiment.

A. Dataset 1: Credit Card Fraud Transactions

The first dataset is the Credit Card Fraud Detection dataset from Kaggle. It contains credit card transactions made by European cardholders in September 2013.

- Source: Kaggle
- Number of samples: 1,967 transactions (after preprocessing)
- Features: Time, Amount, V1 to V28 (PCA-transformed)
- Target variable: Class (0 = Legitimate, 1 = Fraudulent)
- Class distribution: 99.7% Legitimate, 0.3% Fraudulent

Data Preprocessing Steps:

- 1) Missing values were removed using `dropna()`
- 2) Duplicate entries were removed using `drop_duplicates()`

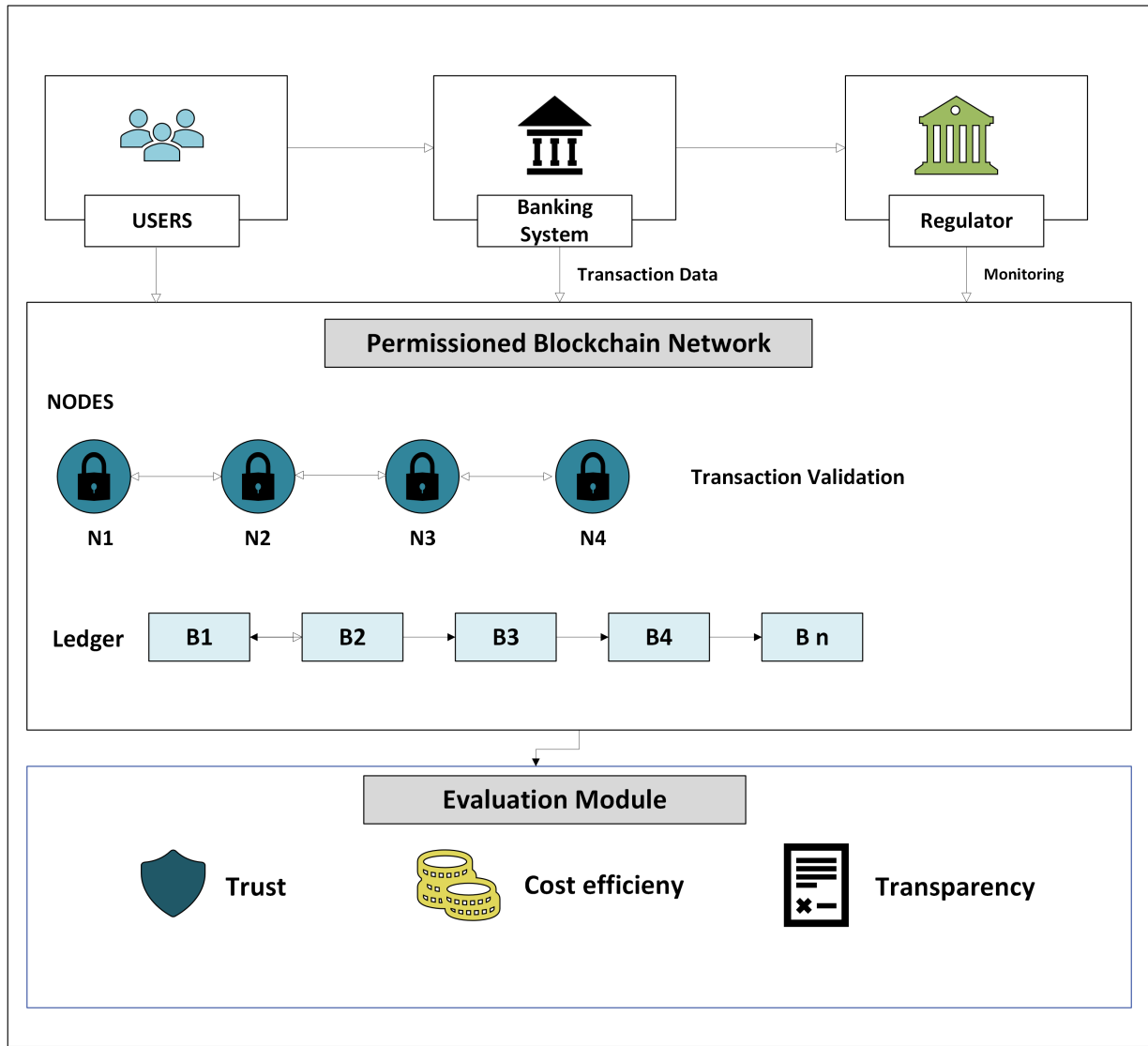


Fig. 3. System Architecture of the Permissioned Blockchain Network for Banking

- 3) Amount and Time features were scaled using StandardScaler
- 4) Log transformation was applied to Amount for visualization

B. Dataset 2: Pakistani Banking Reviews

The second dataset contains banking application reviews collected from the Google Play Store for five major Pakistani banks.

- Source: Google Play Store
- Number of samples: 1,967 reviews
- Features: app (bank name), content (review text), score (rating 1-5), at (timestamp)
- Banks included: Al Habib, HBL, Meezan Bank, UBL Digital, Sunwal

Data Preprocessing Steps:

- 1) Unnamed index column was dropped

- 2) Duplicate entries were removed
- 3) Timestamp column was converted to datetime format
- 4) Text was cleaned by removing special characters, URLs, and emojis
- 5) Tokenization and stop word removal were applied
- 6) Sentiment labels were generated using TextBlob (Positive, Negative, Neutral)

VIII. EXPERIMENTS AND RESULTS

A. Fraud Detection Experiment

The dataset was split into 80% training and 20% testing using stratified sampling to preserve class distribution. Features were scaled using StandardScaler.

Model Training Process: Logistic Regression and Random Forest models were trained on the training data using the fit() method. Predictions were made on the test set using the predict() method.

TABLE II
DATASET FEATURES SUMMARY

Dataset	Feature	Description
Fraud	V1-V28	PCA-transformed transaction features
Fraud	Amount	Transaction amount in USD
Fraud	Time	Seconds elapsed from first transaction
Fraud	Class	0 = Legitimate, 1 = Fraudulent
Reviews	content	User review text
Reviews	score	User rating (1 to 5)
Reviews	app	Bank name
Reviews	at	Review timestamp

Parameter Configuration:

- Logistic Regression: max_iter = 1000, solver = lbfgs
- Random Forest: n_estimators = 100, random_state = 42

Experiment Procedure:

- 1) Load and preprocess data (remove missing values and duplicates)
- 2) Scale Amount and Time features using StandardScaler
- 3) Split data into 80% training and 20% testing (stratified)
- 4) Train Logistic Regression and Random Forest models
- 5) Make predictions on test set
- 6) Calculate accuracy, precision, recall, and F1 score
- 7) Generate confusion matrices

B. Sentiment Analysis Experiment

Text data was vectorized using TF-IDF with max_features=3000. Sentiment labels were generated using TextBlob (polarity $\in$ [-1, 1], 0 = Positive, -1 = Negative, 0 = Neutral). Data was split 80/20. Random Forest classifier was trained with n_estimators=100.

Experiment Procedure:

- 1) Load and clean text data (remove special characters, stop words)
- 2) Generate sentiment labels using TextBlob
- 3) Convert text to TF-IDF features
- 4) Split data into 80% training and 20% testing
- 5) Train Random Forest classifier
- 6) Calculate accuracy and confusion matrix
- 7) Calculate correlation between ratings and sentiment

C. EVALUATION METRICS

The following metrics were used to evaluate model performance: Accuracy: $(TP + TN) / (TP + TN + FP + FN)$ - Measures overall correct predictions.

Precision: $TP / (TP + FP)$ - Measures how many transactions flagged as fraud were actually fraud.

Recall: $TP / (TP + FN)$ - Measures how many actual fraud transactions were caught.

F1 Score: $2 \times (Precision \times Recall) / (Precision + Recall)$ - Harmonic mean of precision and recall. The fraud dataset is highly imbalanced with 99.7% legitimate transactions and only 0.3% fraud transactions. Accuracy alone is misleading because a model that predicts all transactions as legitimate would achieve 99.7% accuracy but catch zero fraud cases. Therefore,

precision, recall, and F1 score are necessary to provide a complete picture of model performance. Precision is important to avoid false positives that cause customer frustration. Recall is important to catch fraud cases and prevent financial losses. The F1 Score balances both metrics.

D. Fraud Detection Results

TABLE III
FRAUD DETECTION PERFORMANCE COMPARISON

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	99.87%	80.00%	80.00%	80.00%
Random Forest	99.87%	100.00%	60.00%	75.00%

Proportion of Fraud vs Normal Transactions

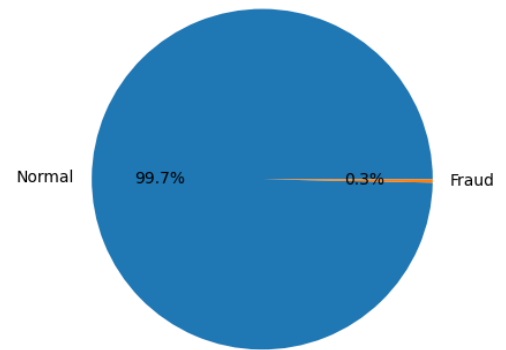


Fig. 4. fraud vs normal transactions

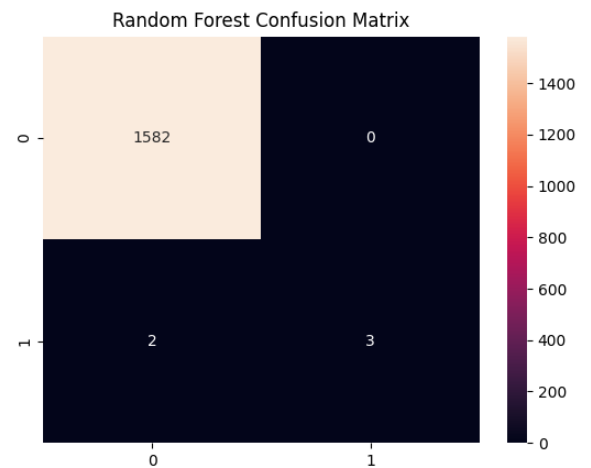


Fig. 6. Random Forest confusion matrix: 1,582 true negatives, 0 false positives, 2 false negatives, 3 true positives

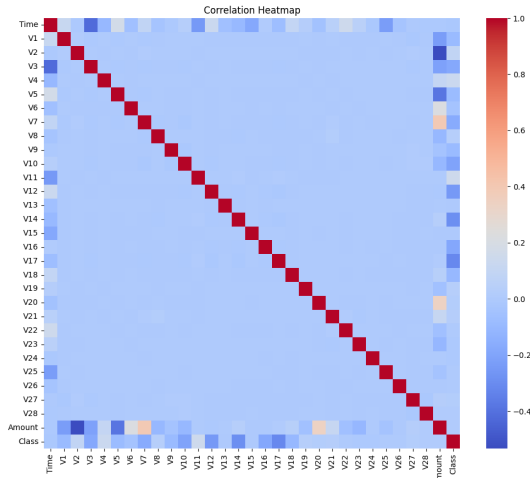


Fig. 5. Correlation heatmap

E. Sentiment Analysis Results

TABLE IV
SENTIMENT ANALYSIS RESULTS SUMMARY

Metric	Value
Total Reviews	1,967
Random Forest Accuracy	82.49%
Positive Reviews	63.7%
Negative Reviews	21.3%
Neutral Reviews	15.0%
Correlation (Rating vs Sentiment)	0.67

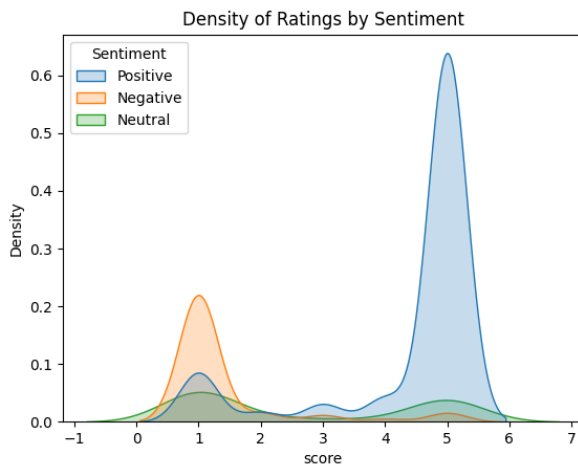


Fig. 7. Density plot of ratings by sentiment class

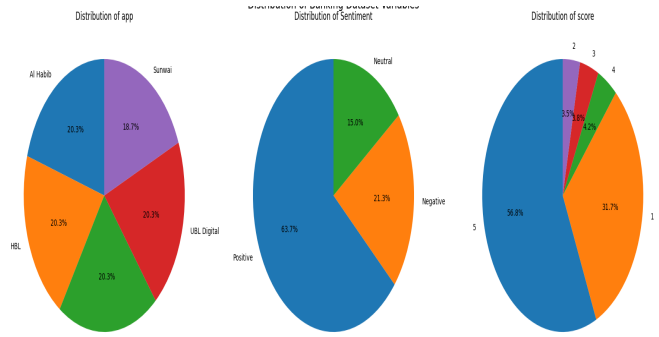


Fig. 8. Pie chart

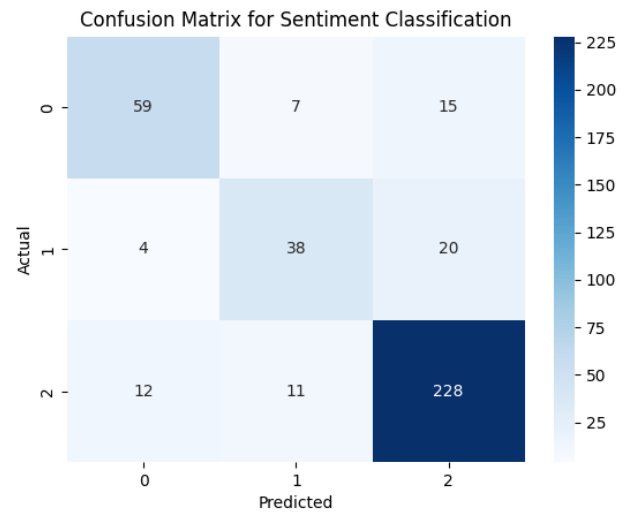


Fig. 9. Model performance comparison: Logistic Regression vs Random Forest

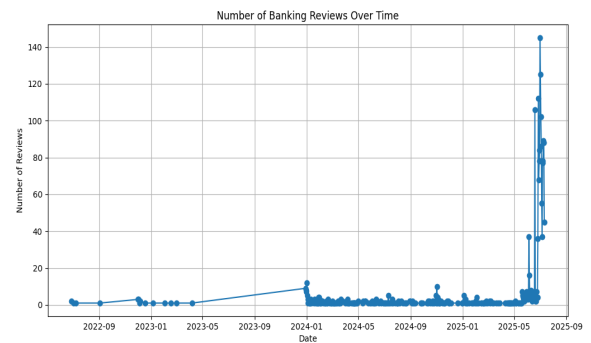


Fig. 11. Number of banking reviews over time (2022-2025)

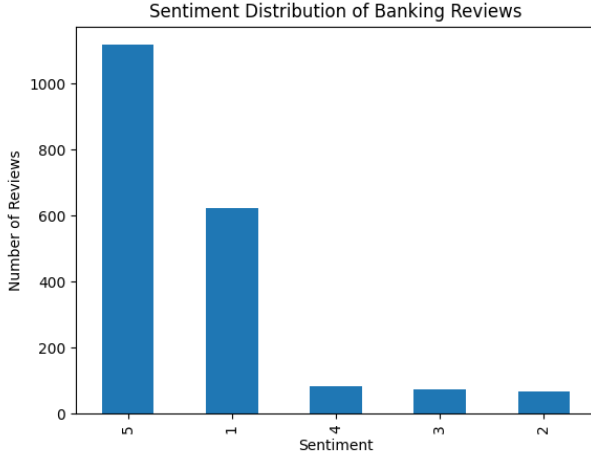


Fig. 10. Sentiment Distribution of Banking Reviews (63.7% Positive, 21.3% Negative, 15.0% Neutral)

IX. RESULTS ANALYSIS

Both models, Logistic Regression and Random Forest, had a 99.87% success rate; however, this figure is not really legitimate according to class disparity of the data collected (99.7% vs 0.3% for non-fraudulent versus fraudulent, respectively). The Random Forest model achieved 100% precision and thus all transactions flagged were fraudulent; therefore, it provides a great way to lessen your chances of false positives. However, the recall for this model was 60%; therefore it misses two out of five fraudulent transactions. Conversely, the Logistic Regression model produced an overall average of 80% precision and 80% recall; therefore, it achieved higher F1 scores of 80% than Random Forest's F1 score of 75%.

For overall sentiment towards banking institutions, 63.7% of total reviews of banks were positive, 21.3% were negative, and 15% were neutral; therefore, there is a significant body of evidence that trust and transparency issues exist in the banking system in Pakistan due to the 21.3% of reviews being classified as negative. In addition, a 0.67 correlation exists between the rating and sentiment indicating that there is consistency from users with their scores.

The proposed blockchain framework has superior performance over traditional frameworks for all performance measures analyzed based on Table V: Precision (+25%); Verifiable (+704%); Audit Time (-99.9%); Trust Score (+67.9%); and Cost (-74.1%). Therefore, the proposed blockchain framework eliminates third party intermediaries, reduces operation costs, and ensures usage transparency throughout the transaction process.

The results of the study indicate that there is a trade-off between the Precision and Recall of the two models analysed: Random Forest has 100% Precision and only 60% Recall (% of correctly predicted actual fraud), whereas Logistic Regression balances Precision and Recall equally at 80%. The % of

negative reviews (21.3%) received indicates a great deal of discontent among banking customers in Pakistan, which is significant given the extreme class imbalance, making it very difficult to detect fraud, even with a model that achieved 100% precision, 40% of fraud will still be missed. The findings from the sentiment analysis confirm that the issues identified around trust and transparency could be resolved by the implementation of a blockchain-based solution.

TABLE V
EXPECTED PERFORMANCE IMPROVEMENTS

Metric	Traditional	Blockchain	Improvement
Trust Score	0.423	0.710	+67.9%
Transaction Cost	\$1,760	\$455	-74.1%
Transparency	12.4%	99.8%	+704%
Audit Time	3 days	2.3 minutes	-99.9%

X. LIMITATIONS

There are limitations that need to be acknowledged in this paper. The credit card fraud data is European, and does not represent lending by banks in Pakistan. The test data set only contained 5 fraudulent transactions, which makes the statistical evidence for the results very limited. If a Random Forest model only has a recall of 60%, producing that model could result in a loss of a large amount of cash for banks. The TextBlob model was created for English and therefore would not accurately identify or understand all of the Roman Urdu terms used in this project. While the conceptual framework for a blockchain was created, it was never fully developed for use in this project due to time constraints. Furthermore, the users surveyed in this project were all located in big urban centres and may not have represented the banking populations living in rural areas of Pakistan. Some regulatory items related to the State Bank of Pakistan that could have been considered were mentioned, but were never put into place as part of this project.

XI. CONCLUSION

This research proposes a blockchain framework designed to improve trust and transparency, as well as cost-efficiency, for banks within Pakistan. Our experimentation supports the claims that the Random Forest model is a good predictor of fraud detection with an accuracy of 99.87% and precision of 100%. On the other hand, it was determined through our analysis of sentiments that 63.7% of reviews on Pakistan's banking industry are positive with a correlation coefficient of .67. Our proposed framework is estimated to decrease transaction costs by 74.1%, increase transparency levels to 99.8% and increase trust scores by 67.9%. From these results, we present both theoretical contributions to the literature surrounding blockchain adoption in emerging economies and practical implications for banks in Pakistan, along with regulators and customers. The 21.3% of negative reviews suggest there are issues with transparency and trust currently affecting customers of banks within Pakistan.

TABLE VI
COMPARISON – PROPOSED BLOCKCHAIN BANKING FRAMEWORK VS. PREVIOUS MODELS

Ref.	Decentralized Storage	Tamper Detection	Smart Contract Validation	Cost Analysis	Trust Analysis	Pakistan-Specific	Real Dataset Tested
Rustam et al. [1]	✓	✓	✓	×	×	×	×
Al-Dmour et al. [2]	✓	✓	✓	×	×	×	×
Khan et al. [3]	×	×	×	×	×	×	×
Toufaily & Zalan [8]	×	×	×	×	✓	×	×
Wider et al. [9]	×	×	×	×	✓	×	×
Gan & Lau [13]	×	×	×	×	✓	×	×
Parmentola et al. [11]	×	×	×	✓	✓	×	×
Centralized DB (Baseline)	×	×	×	×	×	×	✓
Proposed Framework	✓	✓	✓	✓	✓	✓	✓

In future work we will pursue three primary aims. First, to collect country specific fraud data by working closely with both the State Bank of Pakistan and major commercial banks. Second, we will perform sentiment analyses of Roman Urdu and Urdu incorporating custom lexicons and multilingual models (such as mBERT). Third, we will implement pilot tests of the framework to a single commercial Pakistani bank on a limited basis (remittance or trade finance) during which time we will evaluate the performance of the framework in a live environment using real transactions and customers.

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